CHALLENGE PIVOT

GlucoTwin

A patient-specific digital twin for short-horizon glucose-excursion forecasting

Static clinical context + live-like CGM trajectory → updated patient state → 60-minute forecast



Digital Twin Challenge 2026 | Happiest Health

Prepared from supplied Matri-Kavach files + public research.
Model metrics are intentionally not claimed until reproduced.

TEAM: Manas Girdhar (Team Leader) • Moulvi Gufran
COLLEGE / INCUBATOR: [ADD NAME BEFORE SUBMISSION]

The key change: keep the engineering discipline, pivot the clinical use case

Matri-Kavach is valuable work — but its current use case does not satisfy this event's Digital Twin definition.

Current Matri-Kavach

- Maternal emergency referral coordination
- Hospital readiness acknowledgements and handoff closure
- No patient-specific longitudinal physiological state
- No EHR + wearable fusion model predicting a future event

Challenge-aligned GlucoTwin

- Type 2 diabetes: forecast a glucose excursion
- Matched clinical profile + CGM + meal/medication timeline
- Patient state updates when new sensor readings arrive
- Model output, uncertainty, data freshness and clinician view

Decision: submit a separate GlucoTwin concept for this challenge. Keep Matri-Kavach as a separate maternal-health project.

One patient. One defined event. One testable forecast.

A short-horizon forecast is narrow enough to prototype—and specific enough to evaluate.

TARGET POPULATION

Adults with Type 2 diabetes

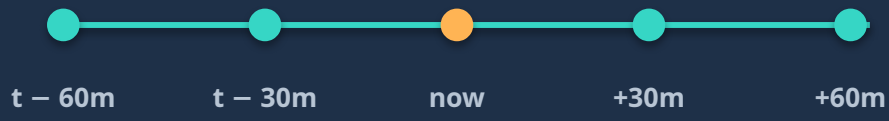
Initial evidence cohort: ShanghaiT2DM research dataset (100 patients; Chinese clinical cohort).

PREDICTION TARGET

Predict whether glucose will cross above 180 mg/dL in the next 60 minutes.

Threshold is a monitoring reference—not a diagnosis, emergency label, or treatment instruction.

FORECAST WINDOW

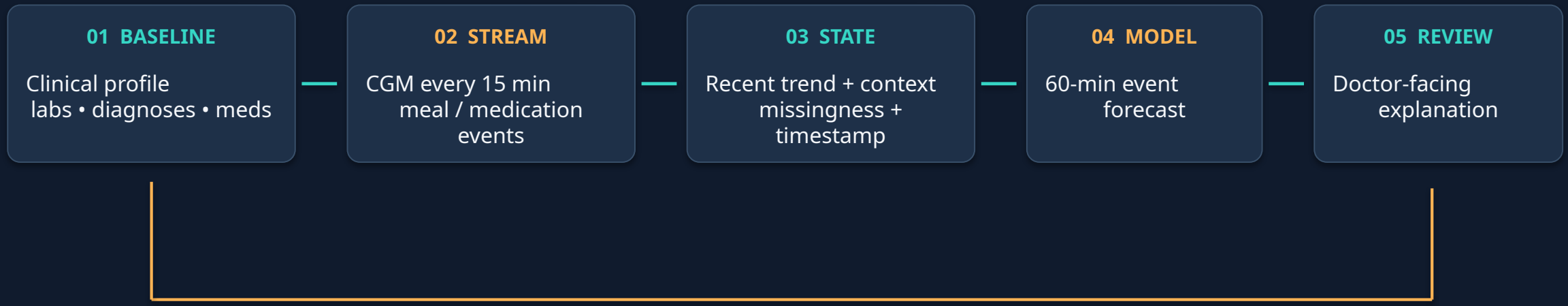


Input: last 4 hours of CGM + baseline clinical features + events recorded before now.

Output: calibrated event probability + freshness / uncertainty state.

A digital twin is an updating patient state—not just a classifier

The virtual state is patient-specific, time-aware, and refreshed as new measurements arrive.



Not a whole-body simulator

A scoped proof of concept.

No automatic treatment

No insulin-dose or medication advice.

Human review remains

Forecast is informational only.

Use real, matched records where possible—never fake a patient match

Primary dataset selected for direct patient-ID linkage between clinical characteristics and CGM.

PRIMARY: ShanghaiT2DM (CC BY 4.0)

- 100 adults with Type 2 diabetes
- 112,475 CGM measurements; 15-minute sampling
- 3–14 days per recording period; 109 T2D files
- Matched clinical/lab profile, medication history and dietary records
- Public Figshare collection; cite the dataset paper

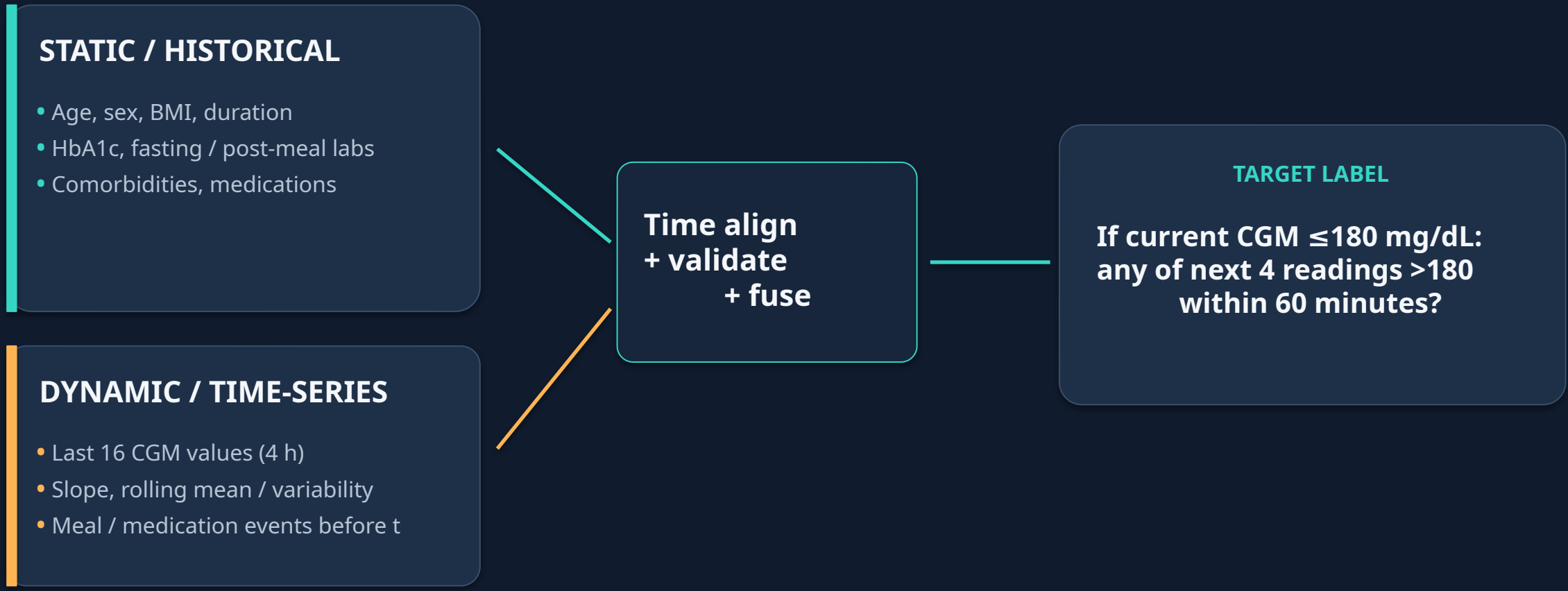
SECONDARY: CGMacros (exploratory)

- 45 participants: 14 T2D, 16 prediabetes, 15 controls
- CGM + Fitbit heart rate / activity + meal macros + baseline labs
- About 10 days; PhysioNet public dataset
- Use only as an independent exploratory check—not merged patient identities
- Small T2D subgroup limits conclusions

Limitations: Shanghai cohort is not an Indian cohort; no claim of Indian-population validation. CGM is the primary wearable/IoT stream.

Exactly how the two streams become one forecast

Fuse only data available at time t; keep future readings strictly in the label.



No cross-dataset patient matching. No future features. Preserve original 15-minute sampling; do not treat interpolated values as independent measurements.

Minimal architecture: reproducible before sophisticated

Start with CSV ingestion + a transparent baseline; add complexity only if it improves held-out performance.

DATA INPUT
ShanghaiT2DM CSVs
clinical + CGM + events

VALIDATE
schema • timestamps
missingness • ranges

STATE / FEATURES
4 h rolling window
static + dynamic fusion

INFERENCE
persistence → LR
→ XGBoost

DOCTOR UI
forecast • reasons
freshness • audit

State store + prediction history

SQLite for PoC; versioned timestamps.

Safety gates

Abstain if stale, incomplete or out-of-range.

Reproducibility

Pinned dependencies, split manifest, tests.

Recommended stack: Python + pandas + scikit-learn / XGBoost + Streamlit + SQLite. FastAPI / React are optional, not MVP requirements.

Model choice is earned by comparison—not assumed

The supplied files contain no trained model or measured performance, so no accuracy is claimed.

BASELINES

- Persistence / current-state rule
- Logistic regression
- Simple CGM-trend model
- Compare before XGBoost

CANDIDATE MODEL

- XGBoost classifier on static + dynamic features
- Probability calibration on validation patients
- Reason codes / feature contribution summary
- Abstain when input quality is insufficient

EVALUATION

- Patient-level 70 / 15 / 15 split
- AUROC + AUPRC; sensitivity / specificity
- Brier score / calibration; false alerts per day
- Ablations: static-only, CGM-only, fused

Critical rule: split by patient ID before training; no patient's windows may leak across train / validation / test.

Doctor view: see the state, the forecast, and its limits

Concept UI only. Replace all illustrative fields with actual application output before recording.

GLUCOTWIN / SYNTHETIC PATIENT SYN-023

DATA FRESHNESS: DEMO ONLY

CURRENT CGM	LAST 60 MIN	60-MIN FORECAST	UNCERTAINTY	LAST SAMPLE
164 mg/dL	Rising trend	NOT RUN	PENDING	DEMO
Synthetic illustrative value	Illustrative only	No model result supplied	Calibration not run	No live feed

ILLUSTRATIVE CGM TRACE — NOT A RESULT



WHY / DATA QUALITY

- Recent CGM slope
- Meal event timestamp
- Missingness / freshness
- Clinician review required

All patient identifiers and values shown here are synthetic placeholders; no trained-model probability is displayed.

The field already exists—our claim must be precise

Existing datasets and forecasting papers make “world’s first” or “no one does this” indefensible.

ShanghaiT2DM (2023)

100 T2D patients; CGM + clinical/lab profile + medications + diet. Dataset, not a complete deployed twin.

CGMacros (2025)

45 participants incl. 14 T2D; dual CGM + Fitbit + meal macros + labs. Rich multimodal data, small cohort.

DiaTrend (2023)

54 people with T1D; 27,561 CGM days and 8,220 pump-data days. Controlled access.

60-min prediction research (2025)

Published models already forecast glucose from CGM; fairness and patient-level generalization matter.

UVA/Padova simulator

Physiological virtual-patient simulation used in artificial-pancreas research; not the same as this T2D task.

A 2026 Scientific Reports study also compares traditional, deep and LLM models on ShanghaiT2DM—so baseline benchmarking and leakage-safe evaluation are essential.

Defensible differentiation: a measurable integration, not a novelty slogan

The potential contribution is a transparent stateful workflow and rigorous evaluation—not a new physiological law.

ESTABLISHED

- CGM forecasting
- Static + time-series feature fusion
- Tree models / recurrent models
- Clinical thresholds and dashboards

PROPOSED GLUCOTWIN BUILD

- Matched patient-level clinical + CGM state
- Meal / medication events time-aligned
- Explicit data-freshness and abstention state
- Patient-level split + fusion ablations

WHAT WOULD PROVE VALUE

- Fused model beats CGM-only baseline
- Calibration acceptable on held-out patients
- False-alert burden is reported
- Results reproducible from public repo

Claim only what experiments show. If fusion does not improve performance, report that honestly and retain the simpler model.

Safety is a product requirement, not a disclaimer

Research prototype only. No autonomous action and no medication or insulin-dose recommendations.

SCOPE

- Predict a future CGM threshold crossing
- Not diagnosis or treatment guidance
- No dose calculation or prescription
- Clinician remains responsible

FAIL-SAFE

- Missing / stale CGM → show warning
- Out-of-range inputs → abstain
- Show forecast horizon and last update
- Log input version and model version

PRIVACY

- Public, anonymized research data only
- Synthetic names in the demo UI
- No patient identifiers in GitHub
- Document dataset and code licences

Predictive performance ≠ retrospective validation ≠ prospective clinical validation ≠ regulatory approval.

What is evidenced—and what is not yet evidenced in the supplied files

This status reflects the uploaded PPT and documents only; update it if code, results or a running demo exist elsewhere.

AVAILABLE FROM UPLOADS

Runnable synthetic-demo source code is now included in the submission package; no ShanghaiT2DM model evaluation or clinical validation is verified.

REQUIRED BEFORE CLAIMING A WORKING TWIN

Load matched ShanghaiT2DM clinical profiles + CGM for real-data evaluation

Do not describe proposed components as implemented or present illustrative traces as model outputs.

Official submission checklist

The official listing requires a public repository and a minimum 20-minute demonstration video.

1–4 full-time students; registered team unchanged	VERIFY
One chronic/lifestyle condition + defined outcome	DESIGNED
Matched static clinical + dynamic CGM streams	DATASET SELECTED
Working model predicts future event	NOT VERIFIED
Doctor-facing conceptual UI	MOCKUP IN DECK
Public GitHub + README required fields	TO BUILD / VERIFY
Open-source licence + dataset attribution	TO ADD
Architecture diagram PDF/PPT	INCLUDED AS SEPARATE PDF
Presentation PDF/PPT	PPTX CREATED
Minimum 20-minute demo video	SCRIPT PROVIDED; RECORDING NEEDED
Prototype/code deadline: 20 Oct, 7 PM IST	URGENT / CHECK PORTAL

Listing inconsistency: top 10 teams appear in the stage timeline; top 5 appear in the Phase 1 description. Confirm with organiser.

Execution order: build the smallest credible end-to-end proof

A short deadline rewards working evidence—not an oversized stack.

P0 • DATA	Download / inspect ShanghaiT2DM; confirm CC BY 4.0 attribution; validate patient IDs and timestamps.
P1 • LABELS	Build 4-hour lookback windows; label future threshold crossing; split by patient before model fitting.
P2 • MODEL	Run persistence + logistic regression + XGBoost; record AUROC/AUPRC, calibration and alert burden.
P3 • UI	Show baseline profile, live-like CGM update, prediction, freshness warning, explanation and audit history.
P4 • RELEASE	Tests, README, public repo, architecture PDF, deck, 20+ minute video, final playback check.

If the model cannot be trained and evaluated before submission, disclose the limitation rather than inventing performance.

Research references and provenance

Official requirements and dataset facts were checked against public sources on 20 October 2026.

Official challenge listing — rules, submission, deadline, video	https://unstop.com/hackathons/crp-digital-twin-challenge-2026-happiest-health-1757873
ShanghaiT2DM dataset paper — Zhao et al., Scientific Data (2023)	https://www.nature.com/articles/s41597-023-01940-7
ShanghaiT1DM / ShanghaiT2DM public data collection — CC BY 4.0	https://figshare.com/collections/Diabetes_Datasets_ShanghaiT1DM_and_ShanghaiT2DM/6310860
CGMacros dataset — Scientific Data (2025)	https://www.nature.com/articles/s41597-025-05851-7
CGMacros dataset on PhysioNet	https://physionet.org/content/cgmacros/1.0.0/
DiaTrend dataset — Scientific Data (2023)	https://www.nature.com/articles/s41597-023-02469-5
Glucose-ML dataset collection and harmonization scripts	https://github.com/Augmented-Health-Lab/Glucose-ML-Project
60-minute CGM forecasting and fairness — PLOS Digital Health (2025)	https://journals.plos.org/digitalhealth/article?id=10.1371/journal.pdig.0000918
UVA/Padova T1D simulator review — Journal of Diabetes Science & Technology (2023)	https://journals.sagepub.com/doi/10.1177/19322968231195081
International Consensus on CGM targets — Diabetes Care (2019)	https://diabetesjournals.org/care/article/42/8/1593/36184/Clinical-Targets-for-Continuous-Glucose-Monitoring
Interpretable glucose forecasting on ShanghaiT2DM (2026)	https://www.nature.com/articles/s41598-025-32373-4

Research claims are paraphrased; see the accompanying research & compliance pack for limitations and source notes.